



Transformative Role of Power Electronics

In solving climate emergency

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Long term shifts in weather patterns and temperature are referred to as climate change. Hotter temperatures, more severe storms, increased droughts, food and water insecurity, and economic disruption, etc. are all due to climate change. According to NASA, “*Human activities (primarily the burning of fossil fuels) have fundamentally increased the concentration of greenhouse gases in Earth’s atmosphere, warming the planet*” [1]. One of the most difficult challenges faced by humanity is to solve the climate emergency [2]. In addition to the climate

emergency, the drive to transition from fossil fuel to renewable energy has taken on a new urgency as a result of Russia's assault on Ukraine [3]. The recent Glasgow summit left a huge hole in addressing the challenges of climate change [4]. In order to avoid more than 1.5°C of eventual warming, the 2030 goal of greenhouse gas (GHG) emissions must be further reduced by approximately 17–20 bn tonnes [4]. Daring and transformative economic transformations are required to save humanity. There are fundamental differences between green and clean energy sources. Free fuel and renewable green energy sources such as solar and wind have negligible negative impact on the environment and require mini-

mum use of water in electric power generation. On the other hand, clean energy source such as nuclear energy does not produce GHG emission during electric power generation, but mining, extraction, and long-term radioactive waste storage are threats to the environment, and Uranium is a nonrenewable resource. In addition, nuclear energy requires massive amount of water in electric power generation. Thus, renewable, and free fuel-based solar and wind power that have minimal negative impact on the environment are considered as green sustainable power. From safety (death rate from accidents and pollution) [5], minimum GHG emissions [5], negligible use of water in electric power generation [6], photovoltaics (PV)

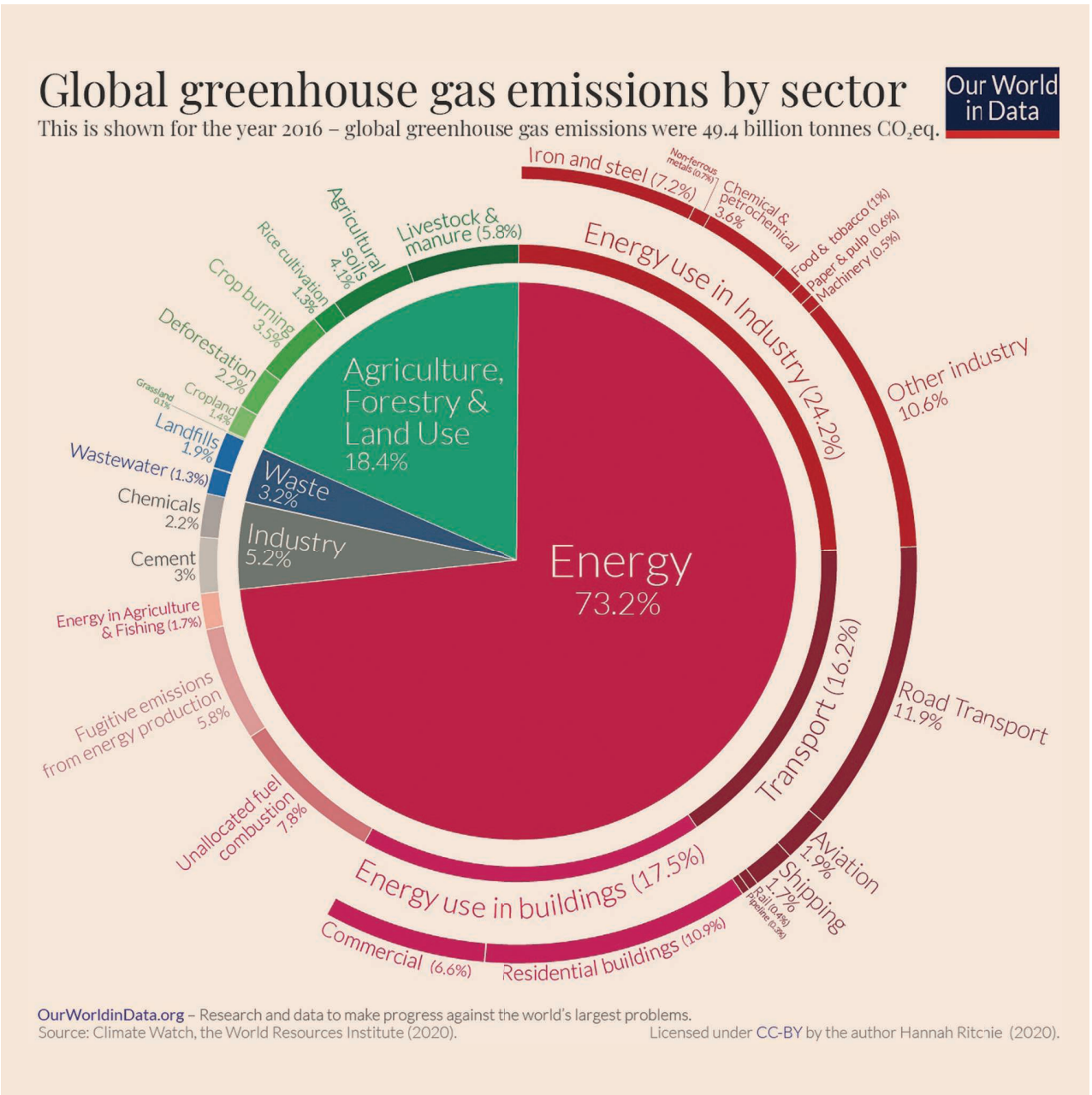


FIG 1 Global GSG Emissions by sectors [9].

and wind power provide lowest cost [5] of electric power generation. In 2020, the International Energy Agency (IEA) announced that solar power is now the cheapest form of energy [7]. The cost of electric power generation by photovoltaics has reached as low as \$0.0104/kWh [8]. The use of these green sustainable energy sources in place of fossil fuels is the only economical solution to address climate emergency. The objective of this article is to show that power electronics has the potential to achieve the goal of electrifying almost everything by green sustainable energy sources and providing a solution to our shared climate emergency.

An Analysis of Greenhouse Gas Emissions

Figure 1 [9] shows global GHG emissions data for various sectors. The livestock-related agriculture emissions can be reduced by making a personal choice either by becoming a vegan or eating synthetic meat. Thus, other than livestock related agriculture emissions, energy is the root cause of all GHG emissions. Recent data have shown that energy use accounts for 83% of the GHG emissions across

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energy and land-use systems [10]. In the following sections we have discussed the role of electrifying almost everything to solve the global climate emergency.

Electrify Almost Everything

The Introduction section has discussed the economic superiority of green sustainable electric power. In addition to the use of green sustainable energy for generating electric power, thermal power must be produced by green electric power. A

number of industries rely on dirty energy sources to generate the required thermal power. GHG emissions by concrete and steel industries account for as much as 16% of humanity’s annual carbon dioxide emissions [11]. As an example, virtually every thermal power step in cement and steel manufacturing can be replaced by green electric power. Figure 2 [12] shows various steps used in cement manufacturing. Other than the use of sustainable green energy for all power needs of manufacturing (including transportation of raw material and finished product), power electronics will play an important role in providing highest energy efficiency for various electrical machineries and equipment used in manufacturing.

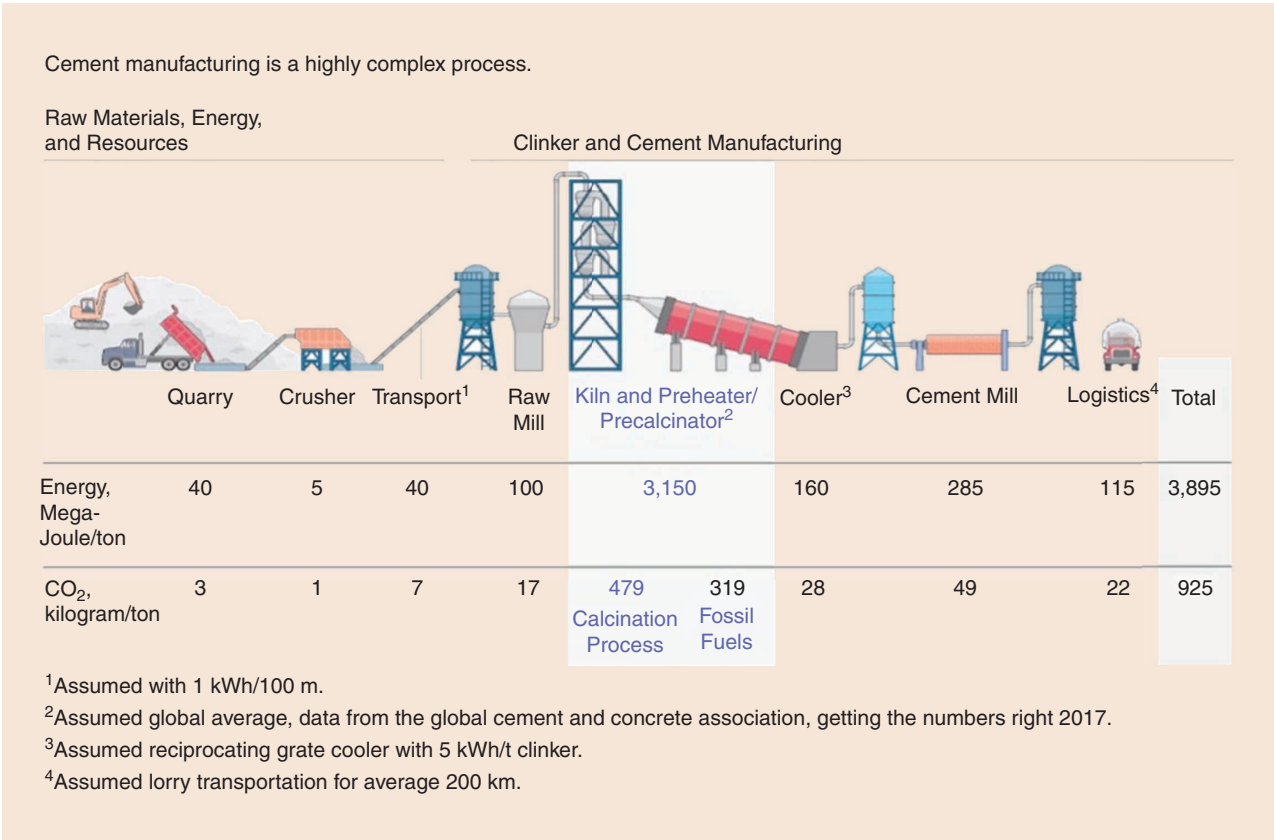


FIG 2 Various steps used in cement manufacturing [12].

Sustainable Sources of Green Electric Power Generation and Storage

Based on abundance, free fuel, minimum amount of GHG emissions, minimum use of water in power generation, highest safety, access to all, and ultra-low cost, solar and wind energy are the only two energy sources for generating green sustainable electric power [13]–[15]. Due to advancements in technology and ultra-large-scale manufacturing, lithium-ion batteries (LIBs) are emerging as a cost-effective solution of electric power storage due to its constantly reducing cost as seen in Figure 3 [16]–[17]. In addition to consumer products and electric vehicles, LIBs are also emerging as a low-cost practical solution at utility scale to address the intermittency of solar and wind energy [14]–[15]. Free fuel-based solar and wind electric power generation combined with batteries for storing electric power has the potential to electrify almost everything.

Myths and Facts about DC vs AC Power

Except for few inductive loads, virtually all our loads (electronics, solid-state lighting, efficient motors etc.) need dc power as the input power. However, based on the outcome of the “war of currents” in the beginning of the last century, the entire world is using ac power-based electricity infrastructure. Power industry has realized the importance of semiconductor-based electronics but has used it only to facilitate the control, conversion, and distribution of ac power. Solar energy-based photovoltaics generates dc power. Batteries and fuel-cells store dc power. Erratic wind power is first converted to dc power before converting to ac power for transmission and distribution. Thus, dc power generated today by free fuel sources is first converted to ac power. After transmission and distribution, the ac power is

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converted back to dc power internally in all dc loads. More than 30% power and capital is wasted in this process [18]. Extremely fast charging of electric vehicles (EVs) requires dc power. As shown in Figure 4 [19], the ac power supplied by the grid is first converted to dc power inside the charger. The components shown in blue color in Figure 4 will not be required if the power source is dc. Another aspect of ac power is that a fortune is invested to regulate the frequency of the ac grid. Thus, from the point of view of capital cost and overall cost of gener-

ated electric power, we need to devise a plan that can be implemented for dc-based electricity infrastructure. However, an immense amount of capital, research, and time has been invested in the current ac-based infrastructure. Thus, we are not proposing to dismantle the existing ac power infrastructure. Other than the existing loads, new loads (e.g., charging of electric vehicles, desalination plants, etc.) are emerging that can be served at a lower cost with a dc power network as compared to the existing ac grid. After retirement of the existing ac infrastructure, new infrastructure can be based on dc power. The power networks architecture proposed in the next section has the potential to transform the global electricity infrastructure and provide a solution to climate emergency by providing efficient dc-dc conversion to achieve an edge over the ac power-based grid.

Power Electronics as an Enabler

The growth of semiconductor-based switching transistors in the integrated circuits (IC) industry has become a boon to the power electronics industry. The availability of high frequency switching transistors has enabled higher efficiency converters to convert dc power to ac power in inverters and transformers. Silicon (Si) based insulated-gate bipolar transistor (IGBT) has been the workhorse of the power electronics industry. The fundamental limits of silicon power electronics can be overcome by the utilization of silicon carbide (SiC) transistors in place of silicon transistors. SiC modules are providing higher power density of motor drive of electric vehicles [20]. The ac power network has sustained so far due to the ability of effective voltage conversion by transformers enabling long distance transmission of ac power with minimal losses. However, due to the advent of SiC power electronics, dc-dc voltage conversion can also be efficiently implemented. The following sub-sections explain our proposed dc power-based network for the most efficient utilization of green sustainable electric power through solar, wind, and batteries. Sub-section A illustrates a local dc power network, and sub-section B includes transmission for a DC network at a larger scale. Key aspects of SiC-based power electronics for high voltage dc-dc conversion are described in sub-section C.

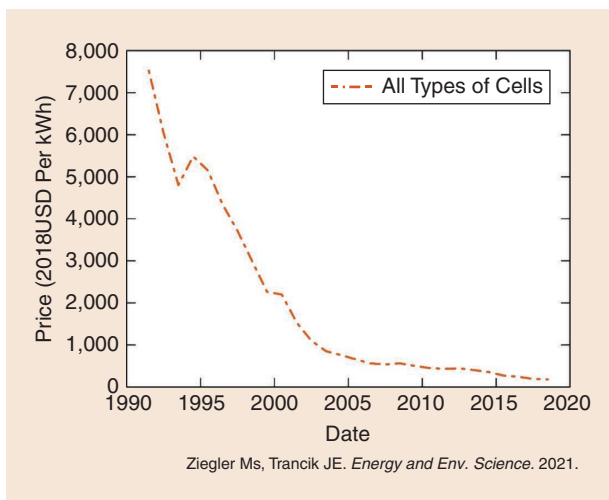


FIG 3 Lithium-ion battery cost reduction [16]–[17].

A. Local DC Power Network

With the advent of solar energy-based PV and wind energy as sources of generation and battery banks for storage, it is possible today to realize Edison’s vision of dc generation and distribution. If the physical distance between power sources (generation and storage) and loads is small (power loss less than about 4–5%), an interconnected

local dc power network can satisfy these loads with maximum efficiency. Figure 5 illustrates the local dc power network.

The first part of Figure 5 is the low-power load (48 V) fulfilment through the PV, wind, and battery bank-based dc network. These loads are typical household appliances with low power consumption like LED lights, ceiling fans,

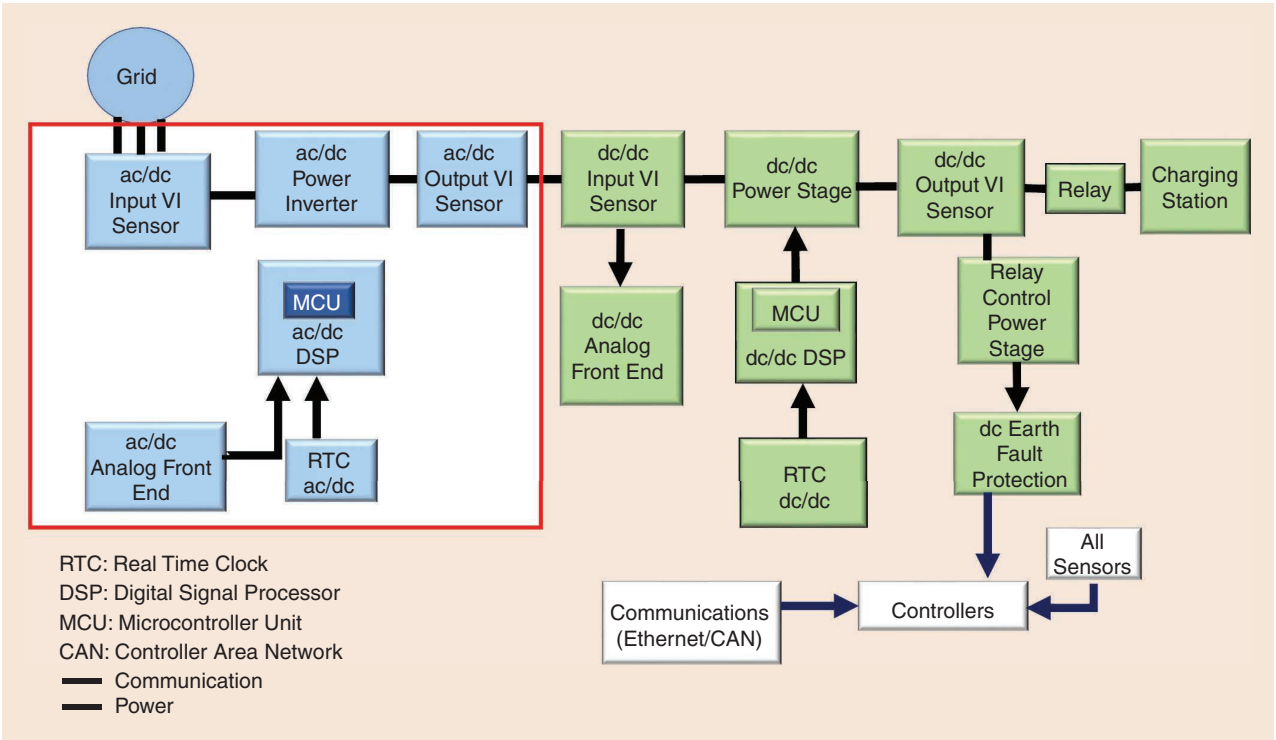


FIG 4 Reduction of capital cost for EV fast chargers if the input power is dc in place of ac [19].

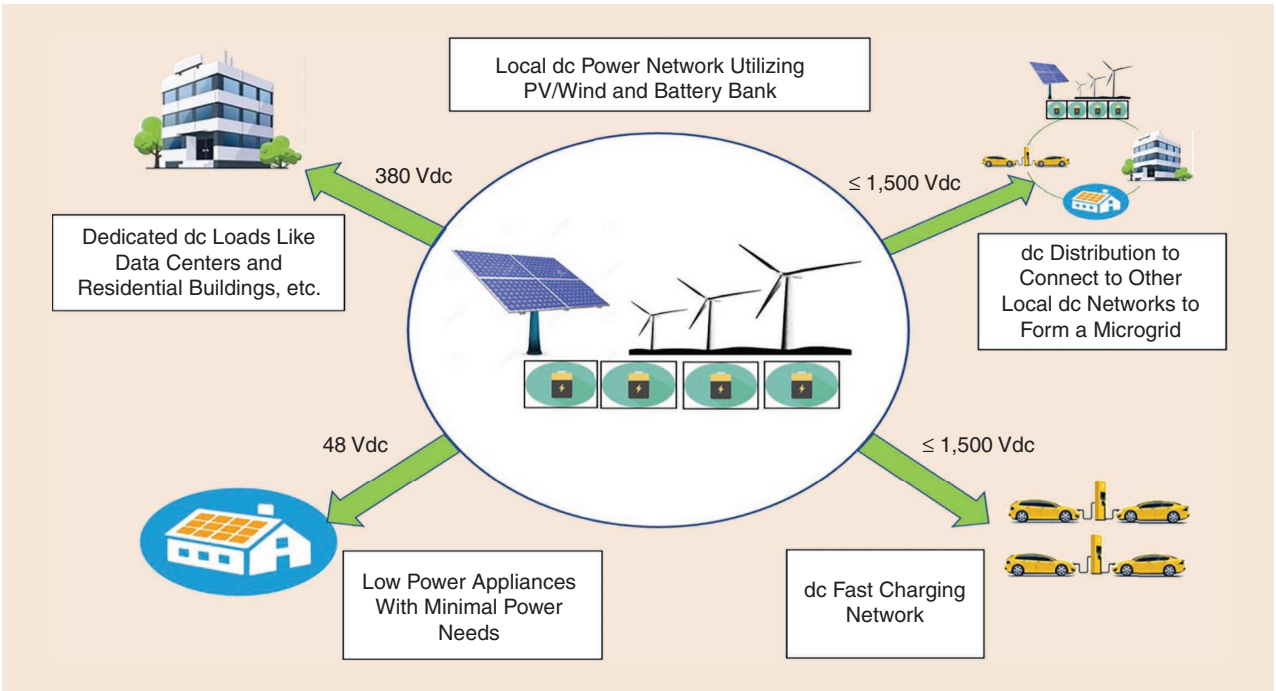


FIG 5 Proposed local dc power network.

etc. The importance of novel dc loads can be seen in reference [21]. The importance of novel dc loads with high energy efficiency must be taken into consideration for under-developed and developing nations where the power infrastructure is still in nascent and developing stages.

The second part of Figure 5 is the medium to high-power loads (380 V) like data centers, commercial centers, and residential buildings, etc. The 380 V dc distribution network is already being used by several data centers in the United States and other countries. However, in most cases the ac power is converted to 380 V dc before entering in the building. The third and most important part is the emergence of dc fast charging networks as loads. With the rapidly growing trend of electrification in the transportation industry, dc fast charging is going to be dominant source of power consumption. The viability of a sustainable dc-based power network for EV charging is elaborated in reference [22]. The last part is the interconnectivity of such a local dc power network with similar local networks to form a microgrid. Such a dc network can be realized only through efficient dc-dc converters which is discussed in the next section.

B. High Voltage DC (HVDC) Power Network for Current and New Electricity Infrastructure

In addition to local dc microgrids, the design for a sustainable power network must encompass the capability to transmit over long distances. Figure 6 illustrates the current transmission network with high voltage ac (HVAC) and HVDC methodologies. The idea of using power networks for sending green sustainable electric power over a distance of thousands of miles has been around for a number of years [23]. Backed by the United Kingdom, India led International Solar Energy Association has a plan to create a network to connect the power markets of 120-plus countries [23]. With

advancements in various aspects of technology, it is possible to realize such power networks today. As an example, at the low-cost of \$0.013/kWh HVDC will be used to connect a 10.5 GW solar/wind complex in Morocco to the UK grid [24]. In current practice, dc power generated from the PV, wind, and battery banks is first converted to medium voltage ac (MVAC) in order to transmit via HVAC. Generation level inverters are utilized for this purpose. The HVDC transmission is carried out utilizing converter stations at the HVAC transmission-level. However, the majority of the system cost in a HVDC transmission system comes from the converter stations. The details of the cost per mile for a HVAC transmission system vs a HVDC transmission system are discussed in reference [14]. For a dc power generation source and a HVDC transmission system, the current practice registers significant losses in conversions.

Our proposed dc network with dc-based power generation and storage and HVDC transmission with new dc loads can be seen in Figure 7. In this system, the generated LVDC power is converted to HVDC power using high efficiency dc-dc converters and transmitted through HVDC transmission. The transmitted dc power can be utilized to power emerging dc loads like EV fast charging and can also serve in the utilization of a dc distribution bus in industrial loads like cement factories, desalination plants, etc. The significance of dc bus in industrial loads like desalination plants is discussed in reference [15]. However, capital and time invested in the current ac infrastructure cannot be ignored just at the cost of energy efficiency. Therefore, the current practice of utilizing a converter station to convert HVDC to HVAC can be incorporated in our proposed network but this reduces the high voltage conversion stage from 2 to 1, thereby reducing converter costs. Existing HVAC transmission lines can be converted to be used with a HVDC

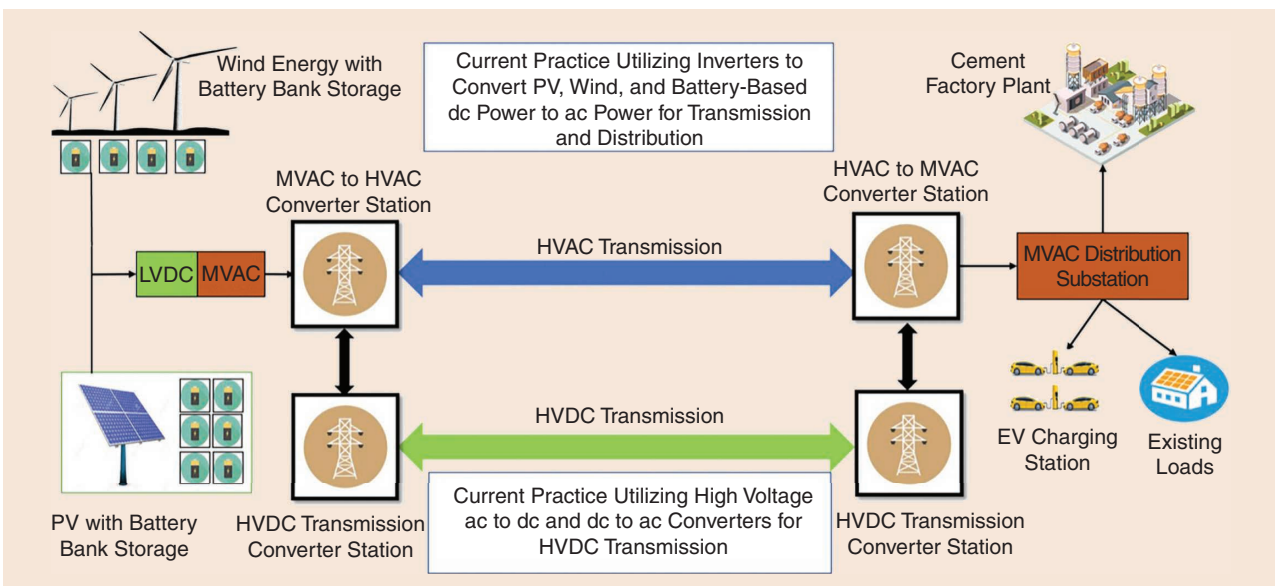


FIG 6 Conceptual diagram representing the current practice of the utilization of green sustainable electric power in the ac power-based grid-tied network.

transmission system, and the new HVDC system would have a higher capacity of electric power transmission than the HVAC system with the same insulation level.

C. Silicon Carbide-based Power Electronics

In both our proposed systems, dc-dc converters play a vital role. As discussed earlier, with the growth and development of power electronics, it is possible to design high efficiency dc-dc converters. The advent of high frequency transformers or solid-state transformers (SSTs) have made it possible to achieve high efficiency ac-ac conversions. Implementation of conversion stages in SSTs utilize power electronics. In our system, the LVDC and MVDC conversions can be handled by Si IGBT transistors as they have adequate performance in the 1–2 kV range. As the breakdown voltage increases, due to limited performance, the highest voltage rating of the state-of-art commercial Si IGBT has been 6.5 kV for the last 15 years. These intrinsic physical limits become a barrier to achieving higher performance power conversion. Therefore, the implementation of HVDC conversion will require transistor ratings that are much higher than commercially available Si IGBT ratings. SiC transistors have a higher breakdown voltage as compared to their Si counterparts. However, Si IGBTs have a significant cost advantage over SiC transistors [25]. Significant manufacturing advances that have been established in the Si industry will be required in the SiC industry as well. Due to the emerging demand for SiC electronics in the power and EV industries, SiC manufac-

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turing can follow a similar roadmap to the Si industry at a much faster rate due to the existing knowledge from the Si based semiconductor industry. Thus, SiC transistors can be potentially cost competitive with Si IGBTs enabling their adoption at a larger scale than the current scenario.

Therefore, SiC transistors are at the heart of our proposed dc-dc power conversion system by utilizing a SST, as shown in Figure 8. There are extensive research activities in the implementation of SSTs for ac-ac, dc-ac or ac-ac conversions [26], [27]. There are

various topologies in consideration for employing the best ac-ac conversion design for grid-based applications based on SSTs. Bidirectional power flow is a key required feature for implementation in our proposed power network. Due to the wide range of dc-dc voltage converters in our proposed system, a highly scalable and modular converter is required. There are various transformer-coupled and direct-coupled conversion topologies in active research. Direct-coupled converter topologies require heavy filtering components and are suitable for MV-to-MV conversions. Transformer-coupled converter topologies are suitable LV-to-MV conversions. We will be extrapolating the transformer-coupled topology for MVDC-to-HVDC conversion to facilitate the end-to-end dc power network illustrated in Figure 7. The topology utilized in our system will be the bidirectional soft switching resonant converter as shown in Figure 8. The modularity of this topology as well as elimination of heavy filtering components reduce system complexity as well. We will be focusing on the power electronics aspect of the

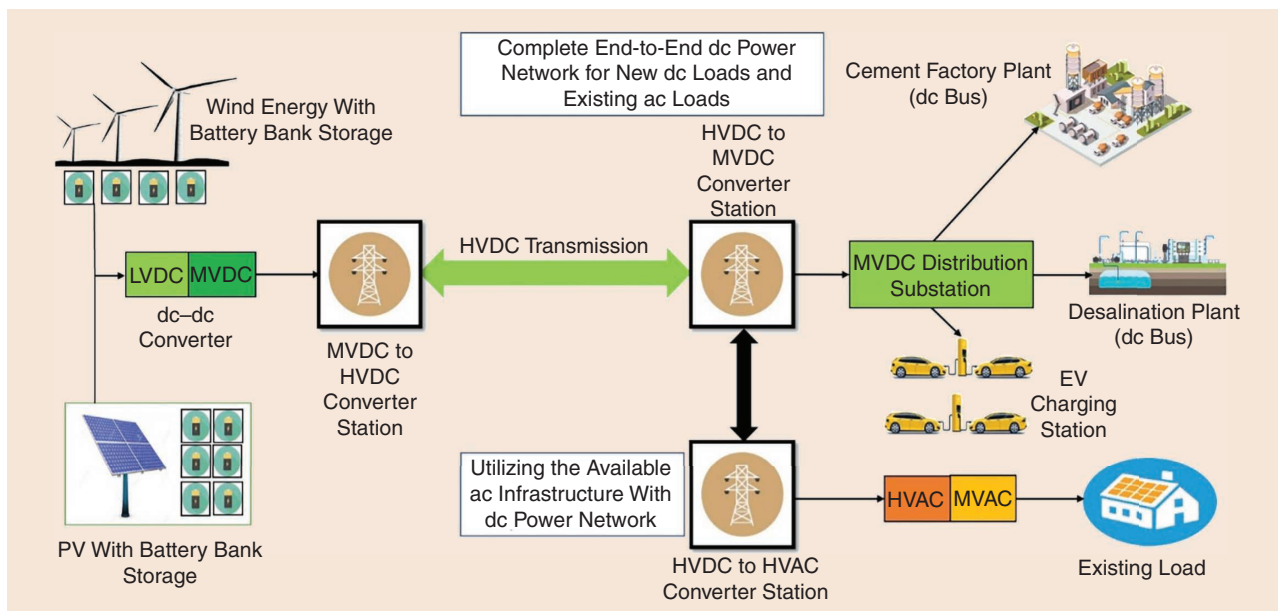


FIG 7 Proposed concept of the end-to-end dc power-based architecture for existing and new loads.

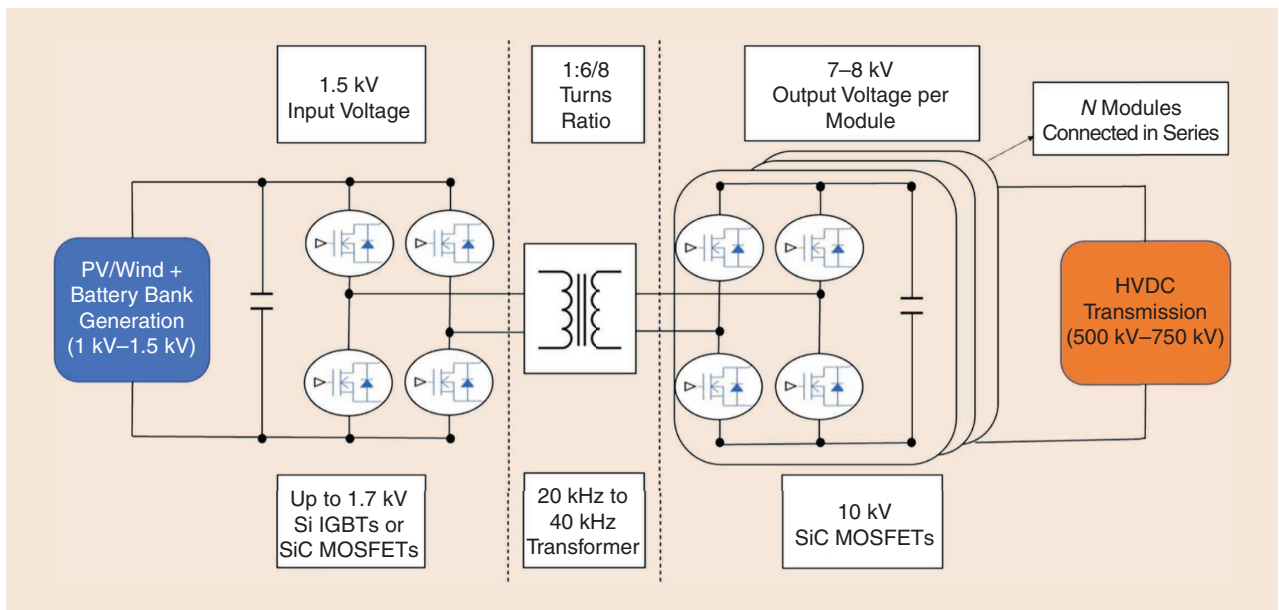


FIG 8 Proposed design of bidirectional soft switching resonant LVDC-HVDC converter.

transformer-coupled topology hereon. As seen in Figure 8, the LVDC (1–2 kV) power generated from green sources undergoes an initial high frequency switching stage in the order of a few kHz. Since this stage faces a LVDC/MVDC input, the switching transistors can be Si IGBTs or SiC transistors depending on the trade study between cost and performance as both transistor types have breakdown voltages up to 1.7 kV. The generated high frequency MVDC output acts as the primary input for a high-frequency transformer (SST) which has superior characteristics in terms of weight, volume, and modularity over the conventional line-frequency transformer. High-frequency suitable magnetic core materials, such as nanocrystalline core, are utilized for this application. The turns ratio can also be designed according to the input voltage from the generation side. The turns ratio cannot be manipulated to achieve HVDC outputs due to the breakdown voltage limitations of the output power transistors. Taking 10 kV SiC MOSFET as an example, the output must be less than 10 kV (~7–8 kV). However, the output power electronics module can be replicated in a series arrangement to achieve the desired HVDC output. The number of SiC power electronics output modules N is directly correlated to the required HVDC output. The modular proposed model will face some challenges at the power electronics level and system level. The details about the power electronics-based challenges like inadequacy of a single device to interface HVDC, switching losses at HVDC operation, device-level isolation and control, high-frequency transformer tradeoff design between insulation and thermal management; and system level challenges like reliability issues, system isolation and dc protection, and system complexity have been extensively discussed [28] and some of them are still ongoing research topics.

Conclusions

Electrification of almost everything by green sustainable electric power is an economic solution to address climate emergency. A large amount of green electric dc power generated by photovoltaics and/or wind turbines and stored in batteries is wasted in current ac electricity infrastructure. Based on silicon carbide power electronics, we propose HVDC transmission of bulk power by dc-dc power converters. Power electronics with higher performance, higher reliability, and low-cost power has the potential to provide transformative transformation of power transmission, distribution, and utilization.

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